

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2025

Course: BTech

Branch: Common to All Branches

Semester: 1st

Subject Code & Name: BTES204 & Computer Programming in C

Max Marks: 60

Date:19/01/2026

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	RAM is a type of a. Permanent memory b. Input device c. Storage device d. Volatile memory	Remember/CO1	1
2	Which topology connects every device to a central hub a. Mesh b. Ring c. Bus d. Star	Apply/CO1	1
3	Which command checks code for errors without running it? a. Run b. Debug c. Compile d. Execute	Understand/CO1	1
4	What is the output: <pre>int main() { int p, q, r; p = 8 > 5 > 2; q = 8 > 5 > 0; r = 8 > 5 > 1; printf("%d %d %d", p, q, r); return 0; }</pre> a. 0 1 1 b. 1 0 0 c. 0 1 0 d. 0 0 1	Apply/CO4	1
5	What does the prefix operator ++i do? a. Increments i and returns old value b. Increments i and returns new value c. Only increments after expression d. Decrements i	Understand/CO3	1

6	What is the data type of character literal 'A'				Understand/CO3	1
	a. char	b. string	c. int	d. unsigned char		
7	The break statement is used to				Remember/CO4	1
	a. Skip one iteration	b. Stop the program	c. Exit from a loop or switch	d. Jump to a label		
8	What will be the output of the following code? <pre>int i = 1; while (i <= 3) { printf("%d ", i); i++; }</pre>				Apply/CO5	1
	a. 1 2	b. 1 2 3	c. 0 1 2 3	d. Infinite loop		
9	Which of the following correctly initializes a character array with a string				Understand/CO4	1
	a. char s[5] = {'H','e','l','l','o'};	b. char s[] = "Hello";	c. char s = "Hello";	d. char s[5] = "Hello";		
10	What will be the output of the following program? <pre>#include <stdio.h> int main() { int a[3] = {1, 2}; printf("%d", a[2]); return 0; }</pre>				Apply/CO5	1
	a. 0	b. 1	c. 2	d. Garbage value		
11	Which of the following correctly declares a pointer to an integer?				Understand/CO4	1
	a. int p;	b. int *p;	c. *int p;	d. pointer int p;		

12	A structure in C is used to:				Understand/CO4	1
	a. Store similar data types	b. Store different data types under one name	c. Store only integers	d. Store only characters		
Q. 2	Solve the following.					12
A)	Explain the switch statement with syntax, rules, and an example.				Understand/CO3	6
B)	Explain the initialization of two-dimensional arrays. How are elements stored in memory? Illustrate with an example.				Remember/CO4	6
Q.3	Solve the following.					12
A)	Write a detailed note on the types of computer software with suitable examples				Remember/CO1	6
B)	What will be output of the following program segment in C and choose the correct option: I) int a =-1, b=1,c,d=2; c= a++ ++b---d; printf("a=%d b=%dc=%d d=%d", c,a,b,d); a) a=0 b=1 c=1 d=1 b) a-1 b=0 c=1 d=1 c) a=0 b=0 c=2 d=1 d) a=0 b=2 c= 0 d=1 II) int main() { char x='X'; char a= ++x+1; printf("%c", a); printf("%d",a); } a) Z 90 b) 90 Z c) error d) none				Apply /CO5	6

	III) <pre>#include<stdio.h> int main() { int a=-2; ++a; if (a) printf("Hi this is CPC "); else printf("C is Programming language "); }</pre> a) Hi this is CPC b) error c) C is Programming language d) None		
Q. 4	Solve Any Two of the following.		12
A)	What happens if the size of an array is not specified during initialization? Explain different cases with examples.	Synthesis/CO4	6
B)	Define simple and compound assignment operators. Give suitable examples for each and explain how they work.	Remember/CO4	6
C)	Design a modular C program that calculates the average of numbers entered by a user, using separate functions for input, processing, and output.	Apply/CO5	6
Q.5	Solve Any Two of the following.		12
A)	1. Define flowchart? draw a flowchart for following algorithm a) Start b) Read the values for B(base) & H (height) c) Calculate area = $1/2 BH$ d) Display area e) Stop	Apply/CO2	6
B)	Design a C program that uses conditional operator to find the largest of three numbers.	Apply/CO6	6

C)	Write a program to demonstrate pointer to integer, float, and character variables.	Apply/CO5	6
Q. 6	Solve Any Two of the following.		12
A)	Write a program to initialize a multidimensional array and calculate the sum of all its elements.	Apply/CO5	6
B)	Define Structure ? Explain how one should access structure fields () explain with suitable program.	Understand/CO5	6
C)	Explain the different types of loops supported in C programming	Remember/CO3	6
	*** End ***		